LABORATORY 04 - APPLICATION LAYER AND PHYSICAL LAYER PROTOCOLS

Julian Felipe Morales Zambrano

Sara Viviana Arteaga Rodríguez

Escuela Colombiana de Ingeniería Julio Garavito

NETWORK ARCHITECTURE AND SERVICES

Bogotá DC, Colombia

2026–1

Objectives

Monitor the application layer protocols.

Review the structured cabling standard and its application.

Perform cable punching with RJ-45 connectors and patch panels.

Tools to Use

Items provided by the university

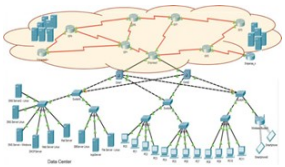
- Computers
- Internet access
- Patch panel and face plate
- Punch tools (for patch cords and impact punches)
- Cable strippers and wire cutters
- Cable tester

Items to be brought by the students

- 4 to 6 meters of UTP/FTP CAT5 or CAT6 cable
- 8 RJ-45 connectors
- If available:
  - Cable stripper or utility knife, and wire cutters
  - Punch tool for patch cords
  - Cable tester

Introduction

We are working on the infrastructure of a company, which typically includes several IT infrastructure services. It consists of both wired and wireless user stations and servers (physical and virtualized), all connected through switches (layer 2 and 3), wireless equipment, and routers that link the network to the internet. It is also common to have cloud infrastructures where resources are provisioned based on the organization's needs. Within the servers, you can find web services, DNS, email, databases, storage, and applications, among others. Let's remember the configuration we are using as the base:



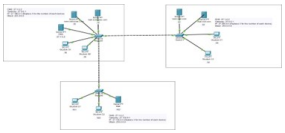
In this lab, we will focus on application layer protocol testing and perform activities related to the physical layer.

Experiments

Understanding how messages circulate across the network and being able to analyze their contents is important for reviewing and fine-tuning the network. In this part of the lab, we will review the information on the application layer and transport layer protocols (only the ports) that we have covered.

Cisco Packet Tracer

[For groups of 1, 2, and 3 students]



Using Packet Tracer, each student should configure the network presented below and document the experience. The services for DNS, HTTP, FTP, and email must be configured on the servers for the network presented.

DIAGRAMA SARA

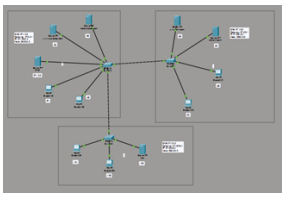
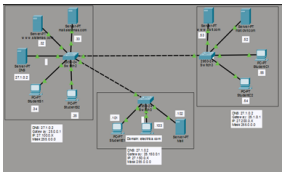


DIAGRAMA JULIAN



Network Configuration

[For groups of 1, 2, and 3 students]

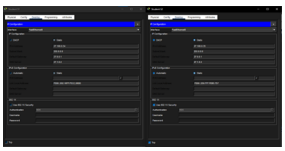
- **En sistemas.com**

Servidor DNS

www.sistemas.com

mail.sistemas.com

StudentS1 y StudentS2



- **En civil.com**

www.civil.com

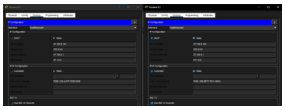
mail.civil.com

StudentC1 y StudentC2

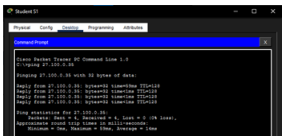
- **En electrica.com**

mail.electrica.com

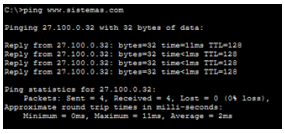
StudentE1 y StudentE2



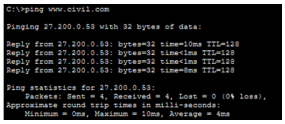
Usando Student S1 PC, mandamos un ping para el Student S2 PC



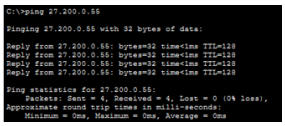
Ahora hacemos un ping a www.sistemas.com



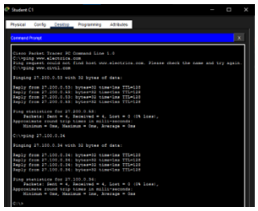
Hacemos la prueba desde el Student S1 a hacer un ping www.civil.com



Hacemos la prueba desde el Student S1 a hacer un ping a Student C1



Ahora desde el Student C1 haremos un ping a Student E1 y probaremos el ping www.civil.com

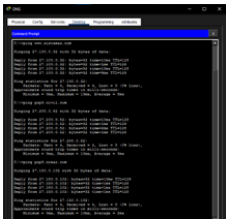


Service Configuration

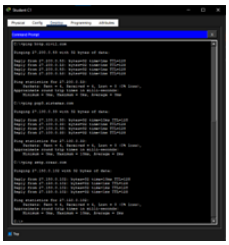
[For groups of 1, 2, and 3 students]

DNS

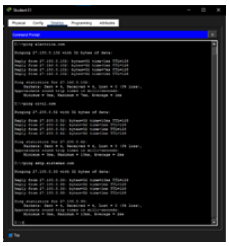
Desde el DNS revisamos cada servicio de cada compañía y hacemos la prueba de algunos de ellos como método de comprobar que si se realizo de manera adecuada.



Haciendo uso del Student C1 probaremos los ping's de tipo http para la compañía civil.com



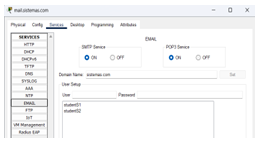
Ahora para la compañía eléctrica.com usando al Student E1



HTTP

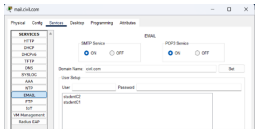
Servidor www.sistemas.com

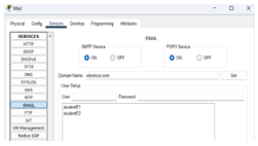
Servidor www.civil.com

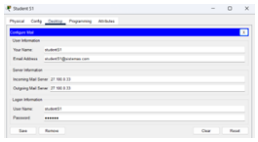


Ahora usamos el modo de simulación y revisamos el contenido PDU de la capa de aplicación (Anexo 22, 23, 24)

Email

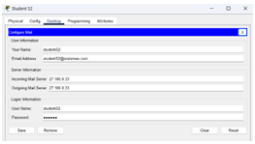


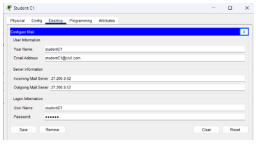




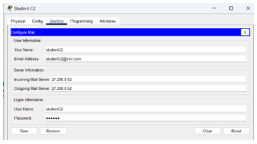
Realizamos un test del servicio, entonces configuramos los correos de los clientes para cada dominio.

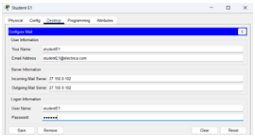
sistemas.com emails:



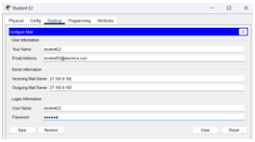


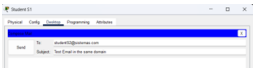
civil.com emails:



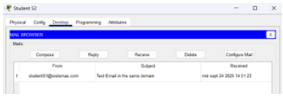


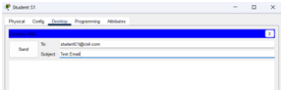
electrica.com emails:



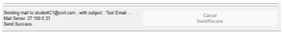


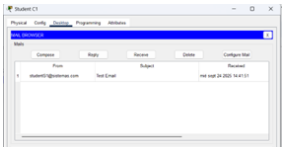
– Enviamos correos entre estaciones en el mismo dominio.



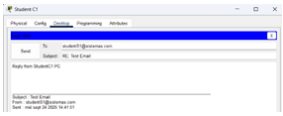


– Enviamos correos a clientes en otros dominios.





– Verificamos cada recepcion de los correos y confirmamos que se recibio.



–Volvemos a revisar con la herramienta de simulacion, los contenido de PDU en la capa de transporte y aplicacion durante el envio de correo enviando procesos entre el cliente que lo envia u el servidor SMTP y entre el cliente que lo recibe y es un servidor POP3. (Anexo 19, 20, 21)

FTP

Configuramos el servicio FTP (PENDIENTE)

In the Real Network

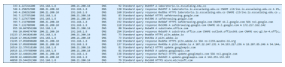
Wireshark

[For groups of 1, 2, and 3 students]

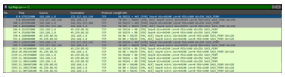
a.

IP: 200.21.200.10 del servidor DNS

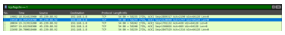
IP: 45.239.88.92 del servidor donde esta alojada



Protocolos:
DNS



La respuesta DNS (Transaction ID 0xbd5e) del servidor 200.21.200.10 resuelve la consulta de laboratorio.is.escuelaing.edu.co indicando que es un alias CNAME que apunta a citrino.is.escuelaing.edu.co, con registro A asignando la IP 45.239.88.92. La zona es gestionada por onix.is.escuelaing.edu.co, confirmando respuesta del DNS institucional sin errores. (Anexo 1)
TCP
Inicio de conexiones



Paquete 1018 (Anexo 2)
El paquete 1018 es el SYN,ACK del servidor (Seq=0, Ack=1) que anuncia ventana 64240, negocia MSS=1440, habilita SACK Permitted, Window Scale=128 y Timestamps. Tras el ACK del cliente se completa el three-way handshake; luego aparecen PSH,ACK con datos HTTP y una redirección 80→443 que inicia el flujo TLS. (Anexo 2)
Cierre de conexiones

Paquete 19410

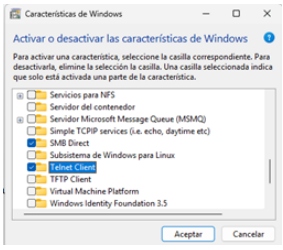
El servidor inicia un cierre ordenado enviando FIN,ACK hacia los puertos efímeros del cliente, terminando múltiples conexiones HTTP keep-alive. El paquete 19410 confirma la terminación limpia (Len=0, Window

Scale=128, Timestamps). La secuencia completa sigue el four-way termination de TCP: cada lado cierra su mitad del canal de forma independiente, sin resets ni retransmisiones.
Anexo 3 problemas en tcp
En la captura se observan Dup ACK, “Previous segment not captured”, Out-of-order y Spurious Retransmission, típicos de Wi-Fi con carga alta. TCP respondió con retransmisión rápida mediante SACK/dup-ACKs, re-sincronizando el flujo sin reset ni caída de sesión.

b.

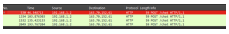
ipconfig /release (Anexo 4 y 5)

Se captura un DHCP Release desde 192.168.1.8 hacia 192.168.1.1 por UDP 68→67, liberando formalmente la concesión actual. Las opciones incluyen 53=Release, 54=Server Identifier y 61=Client Identifier. Este Release dispara la secuencia DORA al ejecutar ipconfig /renew.
Ipconfig /renew (Anexo 6 y 7)



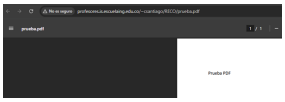
El equipo envía un DHCP Request solicitando conservar la IP 192.168.1.8; el servidor responde con un DHCP ACK entregando Subnet Mask 255.255.255.0, Router, DNS, Lease Time y Broadcast Address por UDP 68→67.

c. Usando TELNET, primero activarlo en nuestro computador:

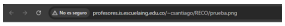


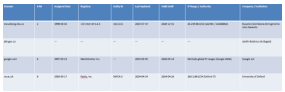
– Capturar los paquetes usando los protocolos TELNET y HTTP:

http://profesores.is.escuelaing.edu.co/csantiago/RECO/index.html
– Telnet (anexos 8, 9, 10, 11, 12)
* telnet profesores.is.escuelaing.edu.co 80
* GET path/file (e.g., GET /index.html).



* Descargar el archivo prueba.pdf, pero no se puede ver correctamente en la consola entonces si queremos verlo tendiramos que guardar el contenido en un archivo pdf.
* Descargar la imagen network.png, pero no se puede descargar porque solo tiene el otro archivo ya visualizado y nos retorna un error 404 Not Found. (Anexo 12)
* Usando la vista de navegador para ver las mismas paginas accedidas con TELNET.





* What difference do you find between the files downloaded via TELNET and via the browser?

La principal diferencia es la forma de visualización con TELNET, para ver cada archivo, tenemos que hacer una copia del contenido descargado y pegarlo en un documento y colocarlo en un formato especifico y asi mismo abrirlo, en cambio en el navegador podemos ver los archivos directamente.

DNS Service Test

[For groups of 1, 2, and 3 students]

Perform the following DNS tests. Go to https://centralops.net/co and check the domains listed below and perform the indicated tests:

escuelaing.edu.co

jbb.gov.co

google.com

Test one more domain from a non-American organization. (Anexo 13, 14, 15)

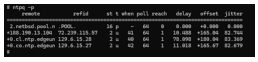
```
# nano /etc/ntp.conf
```

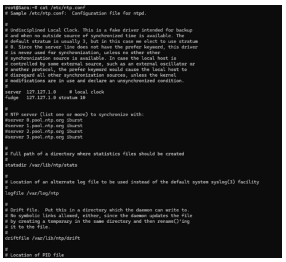
NTP Server

[For groups of 1, 2, and 3 students]

NetBSD

Se instaló ntp con pkgin, se editó /etc/ntp.conf con los servidores de tiempo, se habilitó el inicio automático en rc.conf y se verificó la sincronización con `ntpq -p`.



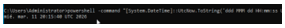


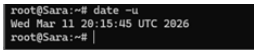
Slackware

Se editó /etc/ntp.conf, se habilitó el servicio con chmod +x y se reinició el demonio. La sincronización fue validada con `ntpq -p`.

Windows Server

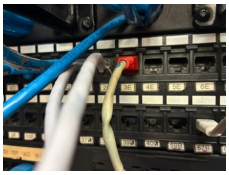
Se usó `w32tm /config` para establecer el servidor de tiempo y net stop/start para recargar el servicio. Se validó con `w32tm /query /status` confirmando ReferenceId correcto y sincronización casi exacta entre máquinas.

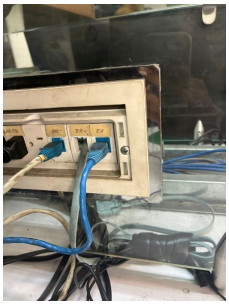


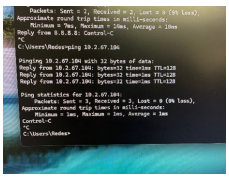


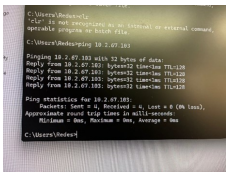
Resultados en los anexos (Anexo 16, 17, 18)

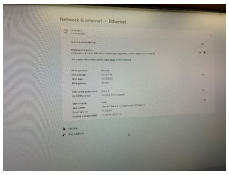
Structured Cabling and Cable Construction

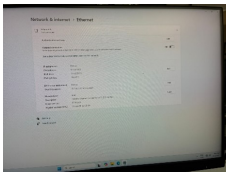












CONCLUSIONES

- El laboratorio permitió entender mejor el funcionamiento de los protocolos de la capa de aplicación como DNS, HTTP y correo electrónico mediante su configuración y prueba en una red simulada.
- El uso de Packet Tracer ayudó a observar cómo se transmiten los datos entre clientes y servidores, identificando los puertos y protocolos utilizados en la comunicación.
- Con Wireshark fue posible analizar el tráfico real de la red y comprender procesos como la resolución DNS y el establecimiento de conexiones TCP.
- El análisis de los paquetes mostró cómo TCP maneja errores de transmisión mediante mecanismos como retransmisiones y ACK duplicados.
- El uso de TELNET permitió visualizar directamente las solicitudes HTTP y entender cómo funcionan las peticiones al servidor.
- Las actividades de cableado estructurado reforzaron la importancia de la infraestructura física para el correcto funcionamiento de una red.

ANEXOS

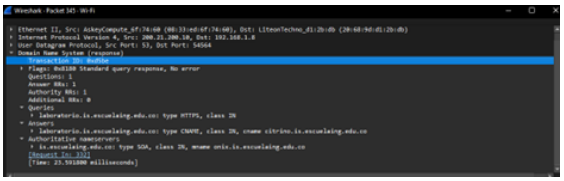


Fig. 1. Anexo 1

Fig. 2. Anexo 2

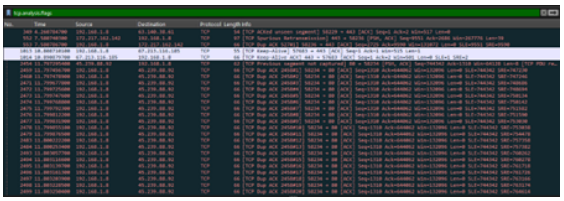


Fig. 3. Anexo 3

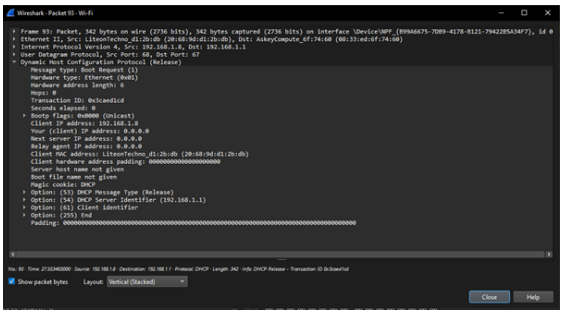
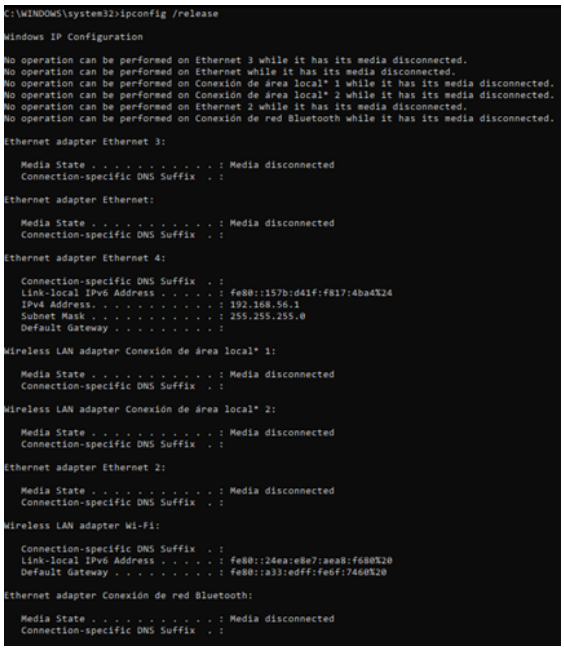
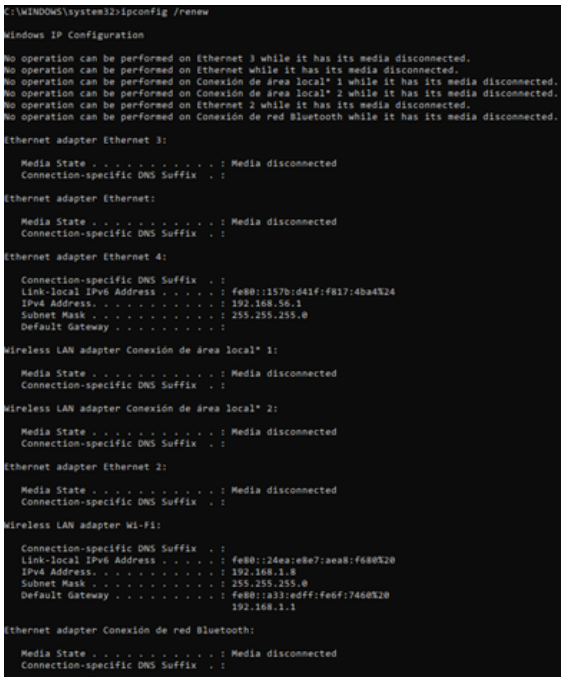


Fig. 4. Anexo 4



```
C:\WINDOWS\system32>ipconfig /release

Windows IP Configuration

No operation can be performed on Ethernet 3 while it has its media disconnected.
No operation can be performed on Ethernet while it has its media disconnected.
No operation can be performed on Conexión de área local* 1 while it has its media disconnected.
No operation can be performed on Conexión de área local* 2 while it has its media disconnected.
No operation can be performed on Ethernet 2 while it has its media disconnected.
No operation can be performed on Conexión de red Bluetooth while it has its media disconnected.

Ethernet adapter Ethernet 3:

   Media State . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet:

   Media State . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 4:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::157b:d41f:f817:4ba4%24
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Conexión de área local* 1:

   Media State . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Conexión de área local* 2:

   Media State . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 2:

   Media State . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::24ea:e8e7:aea8:f680%20
   Default Gateway . . . . . . . . . : fe80::a33:edff:fe6f:7460%20

Ethernet adapter Conexión de red Bluetooth:

   Media State . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
```

Fig. 5. Anexo 5



```
C:\WINDOWS\system32>ipconfig /renew

Windows IP Configuration

No operation can be performed on Ethernet 3 while it has its media disconnected.
No operation can be performed on Ethernet while it has its media disconnected.
No operation can be performed on Conexión de área local* 1 while it has its media disconnected.
No operation can be performed on Conexión de área local* 2 while it has its media disconnected.
No operation can be performed on Ethernet 2 while it has its media disconnected.
No operation can be performed on Conexión de red Bluetooth while it has its media disconnected.

Ethernet adapter Ethernet 3:

   Media State . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet:

   Media State . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 4:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::157b:d41f:f817:4ba4%24
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Conexión de área local* 1:

   Media State . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Conexión de área local* 2:

   Media State . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 2:

   Media State . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::24ea:e8e7:aea8:f680%20
   IPv4 Address. . . . . . . . . . . : 192.168.1.8
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::a33:edff:fe6f:7460%20
                                       192.168.1.1

Ethernet adapter Conexión de red Bluetooth:

   Media State . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
```

Fig. 6. Anexo 6

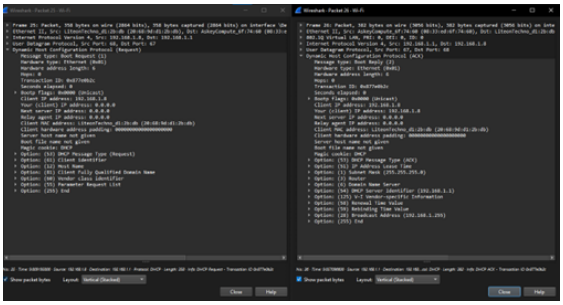


Fig. 7. Anexo 7

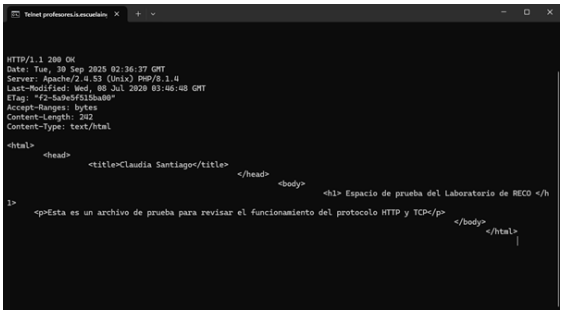


Fig. 8. Anexo 8

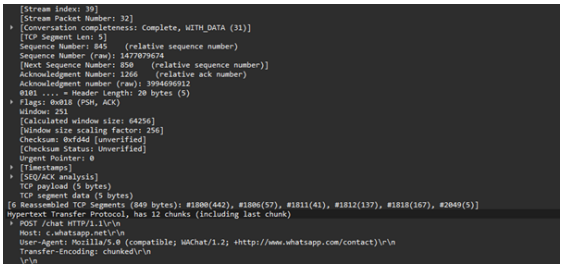


Fig. 9. Anexo 9

Fig. 10. Anexo 10

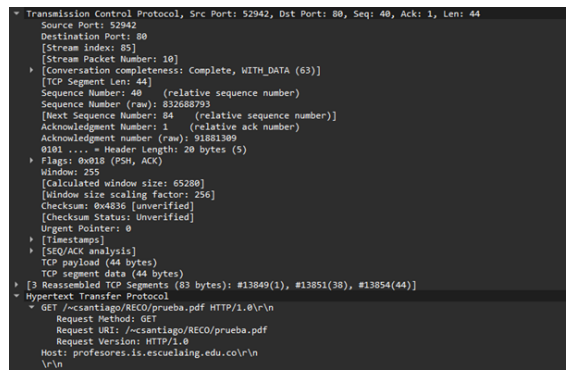


Fig. 11. Anexo 11

Fig. 12. Anexo 12

escuelaing.edu.co IN NS ns2.escuelaing.edu.co
escuelaing.edu.co IN NS ns1.escuelaing.edu.co

Domain Whois record

Queried whois.nic.co with "escuelaing.edu.co"...

```
Domain Name: escuelaing.edu.co
Registry Domain ID: 2586567-CO
Registrar WHOIS Server:
Registrar URL: www.colinternet.com.co
Updated Date: 2022-11-06T18:30:02Z
Creation Date: 1998-06-02T00:00:00Z
Registry Expiry Date: 2025-12-31T23:59:59Z
Registrar: .CO Internet S.A.S.
Registrar IANA ID: 111111
Registrar Abuse Contact Email: soporte@colinternet.com.co
Registrar Abuse Contact Phone: +57.16168941
Domain Status: ok https://icann.org/epp#ok
Registry Registrant ID: REDACTED FOR PRIVACY
Registrant Name: REDACTED FOR PRIVACY
Registrant Organization: Escuela Colombiana de Ingenieria
Registrant Street: REDACTED FOR PRIVACY
Registrant Street: REDACTED FOR PRIVACY
Registrant Street: REDACTED FOR PRIVACY
Registrant City: REDACTED FOR PRIVACY
Registrant State/Province: Bogota
Registrant Postal Code: REDACTED FOR PRIVACY
Registrant Country: CO
Registrant Phone: REDACTED FOR PRIVACY
Registrant Phone Ext: REDACTED FOR PRIVACY
Registrant Fax: REDACTED FOR PRIVACY
Registrant Fax Ext: REDACTED FOR PRIVACY
```

Fig. 13. Anexo 13

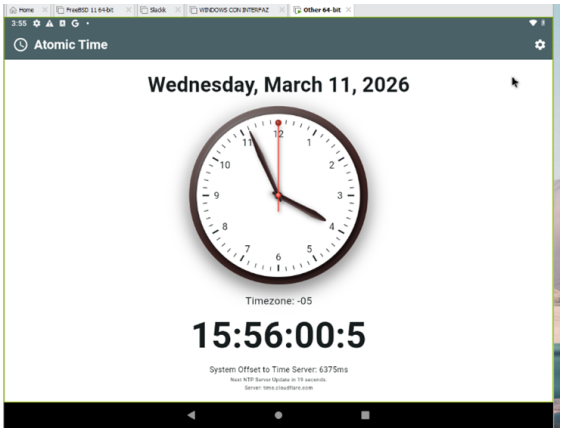


Fig. 16. Anexo 16

```
Domain:
        ox.ac.uk

Registered For:
        University of Oxford

Domain Owner:
        University of Oxford

Registered By:
        University of Oxford

Servers:
        dns0.ox.ac.uk   129.67.1.190
        dns1.ox.ac.uk   129.67.1.191
        dns2.ox.ac.uk   163.1.2.190
        auth4.dns.ox.ac.uk      45.33.127.156
        auth4.dns.ox.ac.uk      2600:3c00:e000:19::1
        auth5.dns.ox.ac.uk      93.93.128.67
        auth5.dns.ox.ac.uk      2a00:1098:0:80:1000::10
        auth6.dns.ox.ac.uk      185.24.221.32
        auth6.dns.ox.ac.uk      2a02:2770:11::21a:4aff:febe:759b
```

Fig. 14. Anexo 14

```
Registrant Contact:
        Domain Registration

Registrant Address:
        IT Services, University of Oxford
        7-19 Banbury Road
        Oxford
        Oxfordshire
        OX2 6NN
        United Kingdom
        +44 1865 273619 (Phone)
        domain-registration@it.ox.ac.uk

Renewal date:
        Friday 26th Jul 2024

Entry updated:
        Friday 10th March 2023

Entry created:
        Wednesday 17th September 2003
```

Fig. 15. Anexo 15

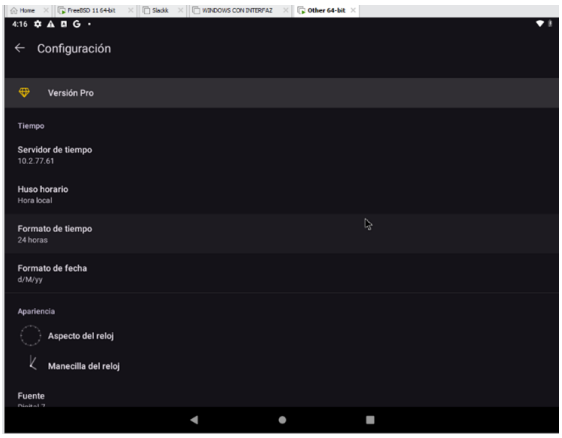


Fig. 17. Anexo 17

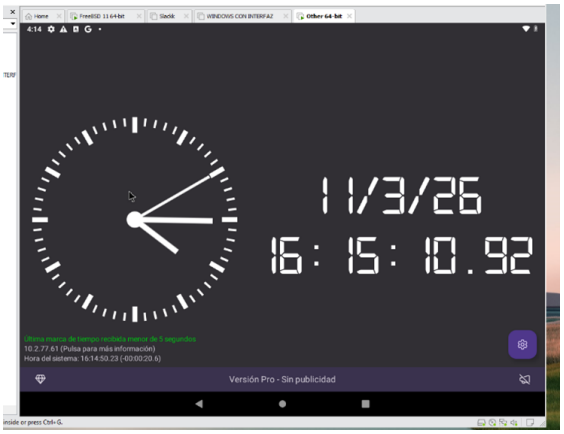


Fig. 18. Anexo 18

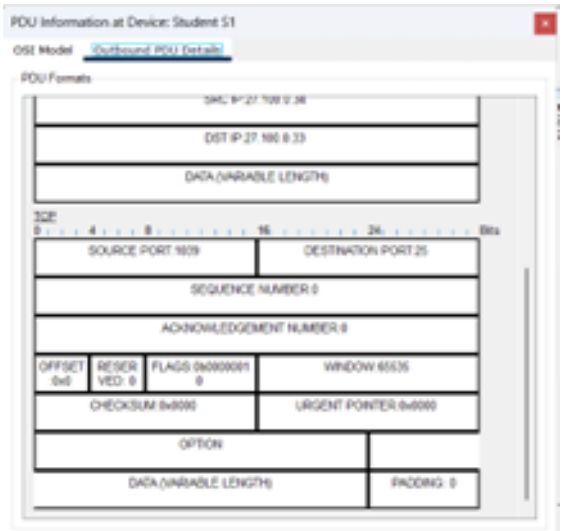


Fig. 19. Anexo 19

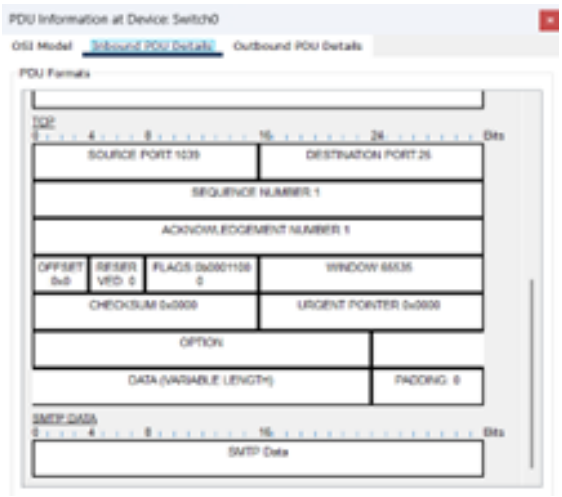


Fig. 20. Anexo 20

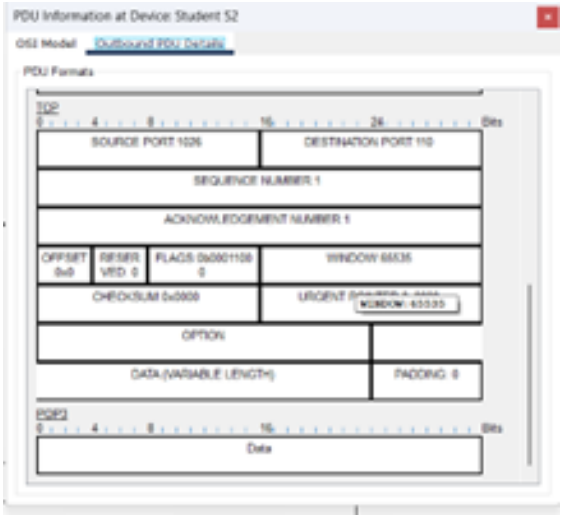


Fig. 21. Anexo 21

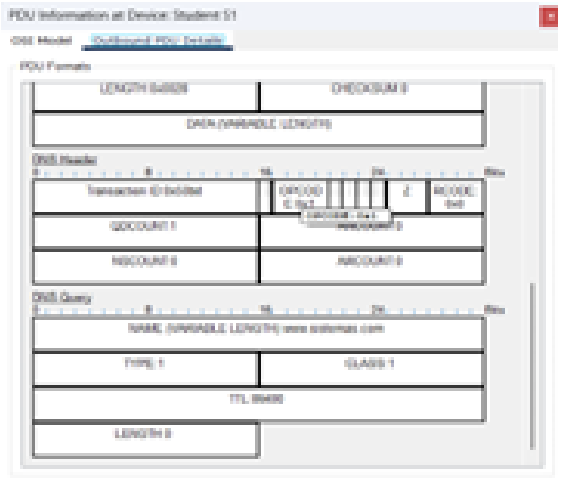


Fig. 22. Anexo 22

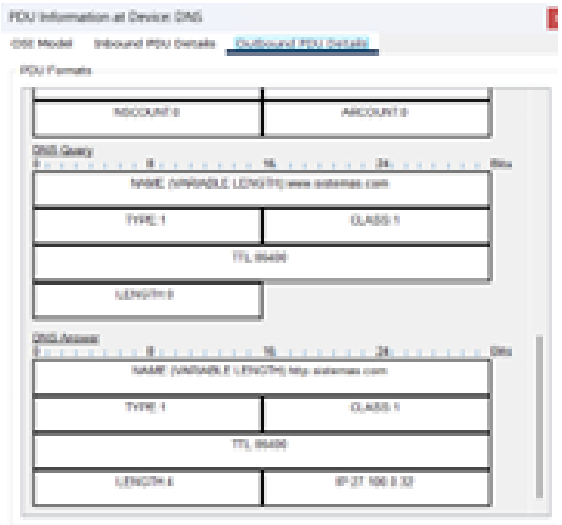


Fig. 23. Anexo 23

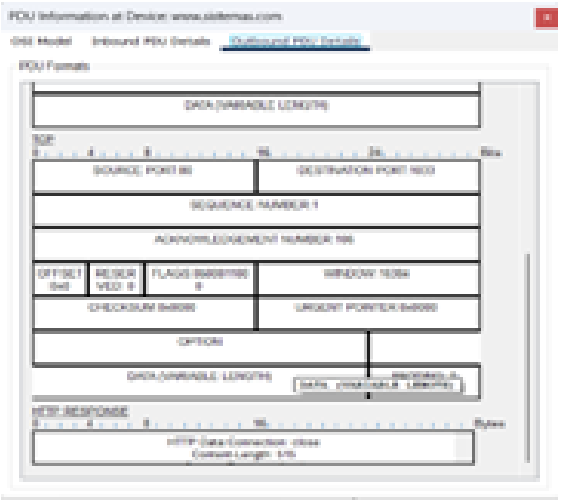


Fig. 24. Anexo 24